* Overloading | Compile time | Static | Same Class
  - (a) Changing the no. of parameters
  - (b) Changing the return type of the parameters
* Overriding | Run time | Dynamic | Multiple Classes

→ Building (P)
(c) ↓ ↓
→ Gym → Restaurant (c)
(a) Override

Abstraction : →

functionality ⇒ Showing what is happening

(Unnecessary) implementation details ⇒ Hiding How it is happening

Hiding the implementation details & only showing the functionality to the "end user" for better User Experience is called data abstraction.

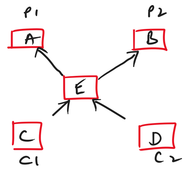
* Abstract Classes & Methods (0 – 100 %)
* Interfaces (100 %)
* If an abstract class has only abstract methods, it is 100 % abstraction.
* If it has even one normal method it is not 100 % abstraction.
* Interface is an entity with completely abstract methods having empty bodies. That's why they exhibit 100 % abstraction.

```
interface interface_name {
    method a();
    method b();
}
```

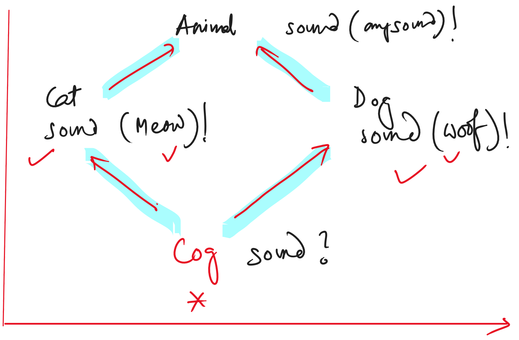
→ These are always empty. They are defined in the child classes.

* Can a class extend multiple classes?
  (a) Y ✗ (b) N ✓

Hybrid Inheritance:
(Interfaces)



Cannot be done using classes

⇒ class E extends A, B {
(Invalid)
}

Diamond Problem:
⇓
Interface
Manual Override